Problem A. Alternating Heights

For a queried subarray, create constraints between adjacent values. We can view the possible heights 1, 2, ..., K as graph vertices. Each adjacent pair adds one directed edge expressing the relative order required by the alternating pattern. The subarray is valid exactly when this directed graph is acyclic: a cycle demands an impossible chain of strict inequalities, while an acyclic graph can be topologically ordered to realize the needed heights.

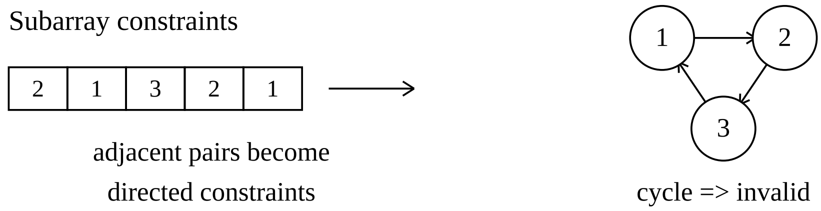

Figure 1. Converting a subarray into a directed constraint graph.

Subtask 1 - binary heights

With only values 1 and 2, validity is the same as ordinary alternation. Mark every position *i* where $A_i = A_{i+1}$. A range [L, R] is valid if and only if there is no marked position between L and R - 1. A prefix sum over these marks answers each query in O(1) after O(N) preprocessing. The total complexity is O(N + Q).

Subtask 2 - preprocess every subarray by graph cycle tests

For each left endpoint L, extend the right endpoint R and insert the next adjacency constraint into the graph on values 1, ..., K. For every candidate subarray, run a cycle test or a topological sort. This directly implements the graph characterization. A simple implementation runs in $O(N^3 + Q)$.

Subtask 3 - handle each query independently

Instead of storing all answers, build the directed graph for the requested range only and test it for a cycle. This is useful when the number of queries is small and avoids large preprocessing memory. The complexity is O(NQ), up to the cost of the small graph cycle checks.

Subtask 4 - monotone invalidity and two pointers

If a subarray is invalid, every larger subarray containing it is also invalid. Therefore, for each L, find the maximum R such that [L, R] is still valid. A two-pointer sweep advances R until a cycle first appears. Then a query [L, R_q] is answered by checking whether R_q is at most this precomputed limit. When the left endpoint moves, rebuild or maintain the constraint graph carefully. The intended bound is $O(N^2 + Q)$.

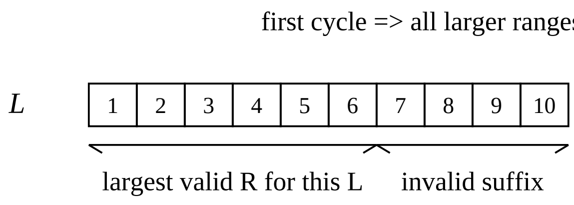

Figure 2. Once a cycle appears, all larger ranges with the same left endpoint are invalid.